Abstract

The automotive and transportation industries are vital to the fast-paced modern world because they facilitate the movement of people and products. The need for precise and effective traffic management and monitoring systems grows as the number of cars on our roadways continues to rise. To address these issues, vehicle number plate recognition (VNPR) technology has become an invaluable resource. Modern traffic management systems, parking lots, and security applications all rely on VNPR because it uses computer vision and machine learning to automatically detect and scan license plates. With a focus on vehicle number plate recognition, this project aims to demonstrate the relevance and promise of computer vision and machine learning for real-world use. The main goal is to create a system that can recognize and record the alphanumeric characters on license plates, automatically identify automobiles, and take pictures of license plates. The goal is to improve automation, security, and traffic management across a range of areas, such as parking management, toll collecting, law enforcement, and the creation of smart cities. The integration of cutting-edge technology, such as artificial intelligence, deep learning, and image processing, is the project's cornerstone. In this project, photos or video feeds from security cameras are captured, the visual data is analyzed, and relevant information is extracted from license plates. In the following sections, we will explore the technical aspects of building a VNPR system, including data acquisition, image analysis, license plates detection, recognizing characters, and system deployment. This project preface acts as an introduction to the idea of VNPR and sets the stage for those sections. We will examine VNPR's theoretical and practical aspects throughout this project, providing a thorough manual for anybody who is interested incomprehending and using this cutting-edge technology.

By the time this project is finished, you will know more about the internal workings of VIN recognition, the difficulties it faces, and its useful uses. Additionally, you will be equipped to make a valuable contribution to the development of intelligent transportation systems and security solutions by having a practical grasp of how to design, create, and implement your own VNPR system.